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Plants being used to cure Diabetes by the local inhabitants of Ranchi District of Jharkhand (India)

Shefa Parveen[a]* & Ashok Kumar Nag[a]
[a]*Department of Botany, Ranchi College, Ranchi, 834001, Jharkhand India

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Abstract : Diabetes is a deadly disease of the Modern world. It shows very little symptom but harms all body parts of humans. Ethnic population due to their changing life style & hereditary reasons are also suffering in great number with this disease. A study was undertaken in Ranchi District of Jharkhand to know about the remedies of Diabetes being in used by the local inhabitants. In the present paper 40 medicinal plants have been enumerated, the way of administration is also described which the local people of Ranchi district mainly the tribes use against Diabetes

Keywords: Diabetes, Patients, Ethnic tribal people, Medicinal values.

INTRODUCTION

Ranchi the capital of Jharkhand presently divided into two subdivisions i.e Ranchi and Bundu. Each sub-division is further divided into blocks, panchayat and villages.There are 14 blocks under Ranchi sub- division and 4 blocks under Bundu sub divisions.The people of Jharkhand include 26.3% of tribal population as per the census report of 2001.These are 30 tribes namely Baiga, Bathaudi, Binjhia, Banjara, Asur, Bhumij, Bedia, Birjia, Chero, Chick, Gond, Oraon, Karmali, Parhaiya, Baraik, Kisan, Khond, Ho, Kharwar, Kharia, Korba, Gorait, Sauria Paharia, Munda, Kora, Lohar, Mahli, Mal Paharia and Savar.

Diabetes(*diabetes mellitus*),[1]is a group of metabolic diseases in which the person has high blood glucose,either because the body's cells donot respond properly to insulin or because of inproper production of insulin.Diabetes patients experience polyuria (frequent urination), polydipsia (increasingly thirsty) and polyphagia (hungry). When body is not able to utilize the hormone insulin properly,person is suffering from diabetes.It can be controlled by proper diet and drugs. Today, diabetes is one of the most common diseases: Type 1 and Type 2 diabetes are two common types of diabetes.Due to lack of information and knowledge of the signs and symptoms most of the patients are unaware of the fact that they are a victims of this disease.The symptoms of the diabetes are common and easy to recognised.There are 8 common symptoms of the diabetes .

1) Unusual increase of thirst and urination, 2) abnormal increase in appetite, 3) mental exertion or fatique, 4)sudden loss in weight, 5) slow rate of healing wounds ,6)dryness of skin,7) blurry vision , 8)tingling or nimbness in hands,legs and feet.In Ranchi many people are suffering from diabetes. According to CPR (crude prevalence rate)around 9% in urban areas and 3 % in rural area patients are there.It has been observed that the highest number of diabetes all over the world is in India.

*Corresponding author :
Phone: 09709113453
E-mail : Shefa.parween@gmail.com

MATERIALAND METHOD

Our survey was carried out in the different remote areas of Jharkhand,specimen have been collected a questionairre was prepared and was given to the patients and the local herbal practitioners. Botanical identification[2,] of these plants[3,4] was done Recent nomenclature of these taxa were confirmed.[5,6,7,] Their medicinal uses [8,9,10] matched with those referred.

Table 1: Details of plants used used by ethnic tribal population.

sl no.	Botanical name &family	Local popular name	Used parts	Method of using
1	*Gymnema sylvestre* R.Br. (Fabacea)	Gurmar	Leaf	Leaf extract
2	*Allium cepa* L. (Alliaceae)	Piyaj	Shoots,Bulb	Orally
3	*Allium sativum* L. (Liliaceae)	Lehsun	Bulb,Compound	Orally
4	*Aloe vera* (L.)Burm.f. (Aloeceae)	Ghrit kumari	Pulp,Slimy inside of the leave	Gel
5	*Azadirachta indica* A.Juss. (Meliaceae)	Neem	Leaf,Seeds	Decoction
6	*Withania somnifera* (L.) Dunal (Solanaceae)	Ashwagandha	Leaves,Roots,Seeds	Decoction
7	*Emblica officinalis* L. (Euphorbiaceae)	Amla	Fruit	Juice
8	*Ocimum sanctum* L. (Lamiaceae)	Tulsi	Whole plant	plant extract
9	*Mangifera indica* L. *(Anacardiaceae)*	Aam	Leaves,Bark,Flowers	Decoction
10	*Catharanthus roseus* (L.) G. Don (Apocynaceae)	Sadabahar	Leaves,Flowers,Root	Plant Extract
11	*Annona squamosa* L., Sp. (Annonaceae)	Sharifa	Seeds,Leaves,Bark	Orally,Decoction
12	*Punica granatum* L. (Lythraceae)	Anar	Seeds,Flowers	Raw
13	*Aegle marmelos* (L.) Correa (Rutaceae)	Bael	Fruit,Leaves,Bark	Juice,Raw
14	*Areca catechu* L. (Arecaceae)	Supari	Fruit	Aqueaus extract
15	*Momordica charantia* L. (Cucurbitaceae)	Karela	Fruit	Juice
16	*Andrographis paniculata* (Burm. f.) Wall.ex Nees (Acanthaceae)	Kalmegh	Whole plant	Plant extract
17	*Syzygium cumini* (L.) Skeels (Myrtaceae)	Jamun	Seeds,Barks,Leaves	Fruit,Juice
18	*Trigonella foenum- graecum* L. (Leguminosae)	Methi	Leaves,Seeds	Aqueaus extract
19	*Ficus benghalensis* L. (Moraceae)	Banyan	Bark,Root-Fiber,Leaves,Seeds	Milky juice i.e Latex
20	*Artocarpus heterophyllus* Lam. (Moraceae)	Kathal	Seeds,Fruit,Leaves	Decoction
21	*Brassica oleracea* L. (Brassicaceae)	Bandha gobhi	Flower	Raw
22	*Ananas comosus* L. Merr (Bromeliaceae)	Ananas	Fruit,Stem	Juice
23	*Phaseolus vulgaris* L. (Fabaceae)	Bean	Green pods	Orally
24	*Zea mays* L. (Poacea)	Makai	Stigmas,Styles	Boil
25	*Citrus limon* L. Burm. f. (Rutaceae)	Nimboo	Fruit,Peel	Fruit
26	*Swertia chirata* L .(Gentianaceae)	Chiraita	Whole plant	Plant extract,Fruit
27	*Camellia sinensis* (L.) Kuntze (Theaceae)	Chai	Leaves,Buds	Aqueaus extract
28	*Ipomea batatas* (L.) Lam. *(Convolvulaceae)*	Shakarkand	Fruit	Orally
29	*Citrus aurantium* (L.) Osbeck (Rutaceae)	Bitter orange	Pulp,Peels,Seeds	Orally
30	*Murraya koenigii* (L.) Sprengel (Rutaceae)	curry patta	Leaf	Boil

Table 1continued.....

Table 1continued.....

31	*Avena sativa* L.(Poaceae)	Oat	Seed	Stalk
32	*Capsicum minimum* L. (Solanaceae)	Mirchi	Fruit,Ripe and Dried	Decoction
33	*Daucus carota* L. (Apiaceae)	Gajar	Fruit	Orally
34	*Carica papaya* L. (Cucurbitaceae)	Papita	Seeds,Fruit	Orally
35	*Malus domestica* Borkh. (Rosaceae)	Apple	Fruit,Flowers,Bark	Orally
36	*Helianthus annuus L.*(Asteraceae)	Suraj mukhi	Seeds,flowers	Oil
37	*Piper nigrum* L. (Piperaceae)	Kali mirch	Dried peppercorns	Orally
38	*Zingiber officinale* Roscoe (Zingiberaceae)	Adrak	Fresh and Dried Rhizomes	Extract
39	*Laurus nobilis L.* (Lauraceae)	Tej patta	Leaves,Fruit,Oil	Decoction
40	*Hibiscus rosa-sinensis* L. (Malvaceae)	China rose	Whole plant	Extract

RESULT AND DISCUSSION

Diabetes, a epidemic type disease is now spreading fastly among the ethnic population too, the reason being the change in life style & genetic reasons. In the present paper 40 different species of plants for getting relief from diabetes were diagonised through survey work among these all the body parts of two species are used.

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